SIMULATION TASK

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CODE:

```
 clc;
clear;
close;

//--------------------------------------------------
// Plant and PID Controller
//--------------------------------------------------
s = poly(0,'s');

G = syslin('c',10/((s+1)*(s+3)*(s+5)));
C = syslin('c',2*(s^2+3*s+2)/s);

// Open-loop transfer function
L = C*G;

// Closed-loop transfer function
T = L/(1+L);

//--------------------------------------------------
// (a) Bode Plot
//--------------------------------------------------
figure(1);
bode(L);
title("Bode Plot of Open Loop");

//--------------------------------------------------
// (b) Nyquist Plot
//--------------------------------------------------
figure(2);
nyquist(L);
```

```
title("Nyquist Plot of Open Loop");

//--------------------------------------------------
// (c) Closed-loop Step Response
//--------------------------------------------------
t = 0:0.01:10;

[y,x] = csim('step',t,T);

figure(3);
plot(t,y,'b','LineWidth',2);
xlabel("Time (sec)");
ylabel("Amplitude");
title("Closed Loop Step Response");
xgrid();

//--------------------------------------------------
// (d) Pole-Zero Map
//--------------------------------------------------
figure(4);
plzr(T);
title("Pole-Zero Map");

//--------------------------------------------------
// Gain Margin and Phase Margin
//--------------------------------------------------
[gm, pm, wg, wp] = g_margin(L);

//--------------------------------------------------
// Step Response Specifications
//--------------------------------------------------
final_value = y($);

// Maximum value
peak = max(y);

// Percentage Overshoot
overshoot = ((peak-final_value)/final_value)*100;

// 2% Settling Time
```

```
upper = final_value*1.02;
lower = final_value*0.98;

settling_time = t($);

for i=1:length(y)
    if and(y(i:$)>=lower) & and(y(i:$)<=upper) then
        settling_time = t(i);
        break;
    end
end

//--------------------------------------------------
// Display Results
//--------------------------------------------------
disp("================================================");
disp("        CLOSED LOOP PERFORMANCE");
disp("================================================");

mprintf("\n%-25s %10.3f dB\n","Gain Margin",gm);
mprintf("%-25s %10.3f deg\n","Phase Margin",pm);
mprintf("%-25s %10.3f %%\n","Overshoot",overshoot);
mprintf("%-25s %10.3f sec\n","Settling Time",settling_time);

disp("================================================");
```

Output: